Why is the estimation of metaorder impact with public market data so challenging?

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Based on:
MN, G. Bormetti, F. Campigli, G. Rodikov, F. Lillo, **arXiv:2501.17096**

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Modeling Price and Trades

Modeling the joint dynamics of price and trades is useful for two main reasons

Forecasting

- Trading
- Risk Management
- Policy implementation
- Forecasting transaction cost

Reproduce market stylized facts

- Build Simulators
- Estimate market impact
- Understand market dynamics

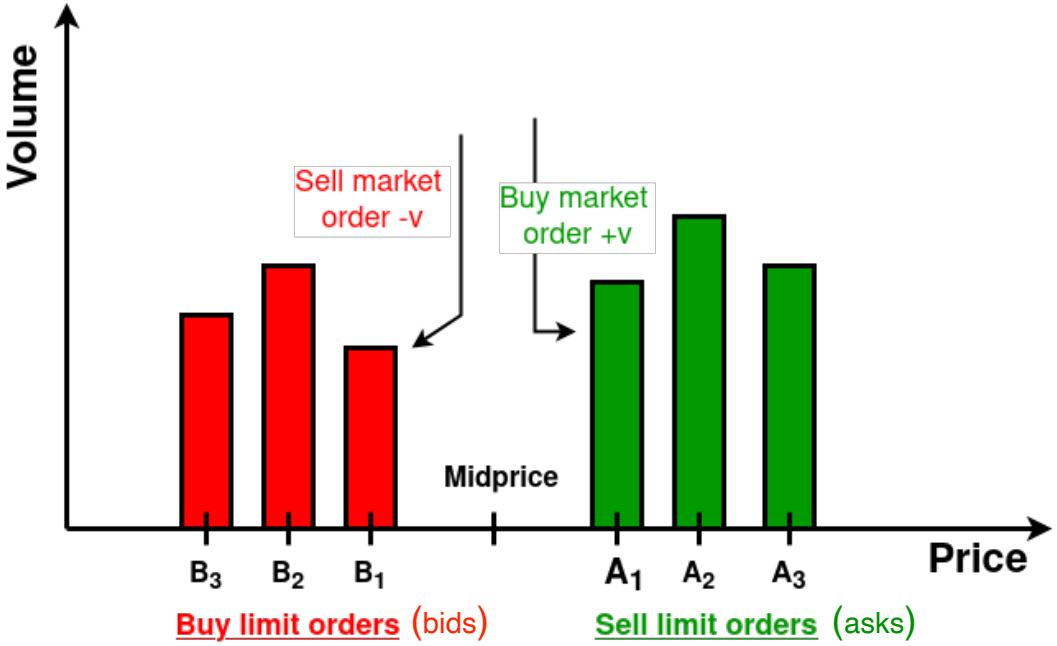
The Limit Order Book (LOB)

The **LOB** is an electronic system that matches customers orders (bid and asks).

Three main actions:
- Limit orders;
- Cancellations of Limit Orders;
- Market orders.

Positive volumes correspond to **buy** orders and **negative** for **seller**.

Price formation is deeply related to volume dynamics.



VAR

VAR framework proposed by Hasbrouck [1]

p = lag order

White noise

$$r_t = \sum_{i=1}^{p} a_i r_{t-i} + \sum_{i=0}^{p} b_i v_{t-i} + u_{1,t},$$

$$v_t = \sum_{i=1}^{p} c_i r_{t-i} + \sum_{i=1}^{p} d_i v_{t-i} + u_{2,t}$$

b_0 measures the contemporaneous impact.

A particular case is the **Transient Impact Model (TIM)** [2] for $a_i = c_i = 0,\ \forall i$ and $p \to \infty$.

The question is whether this model is able to reproduce price reaction to order executions and how we make the simulation once it is calibrated.

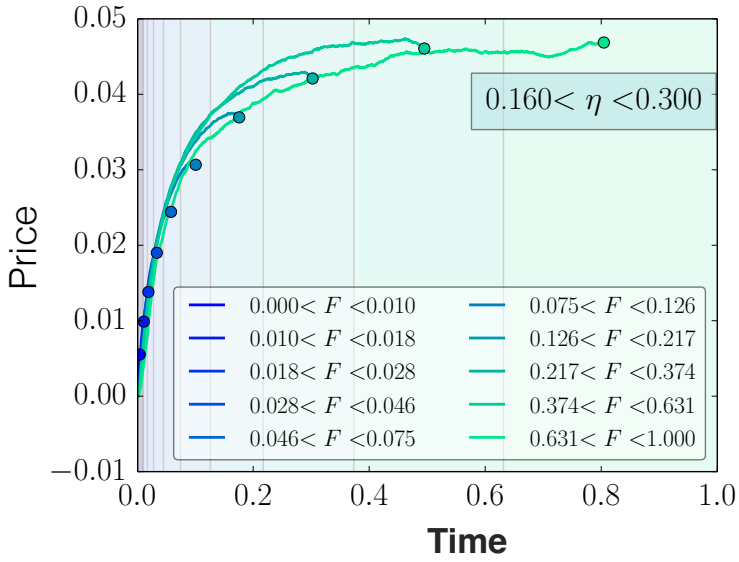
[1] Joel Hasbrouck. The Review of Financial Studies, 4(3):571–595, 1991.
[2] Bouchaud, J.-P., Gefen, Y., Potters, M., and Wyart, M., Quantitative Finance, 4(2):176-190.

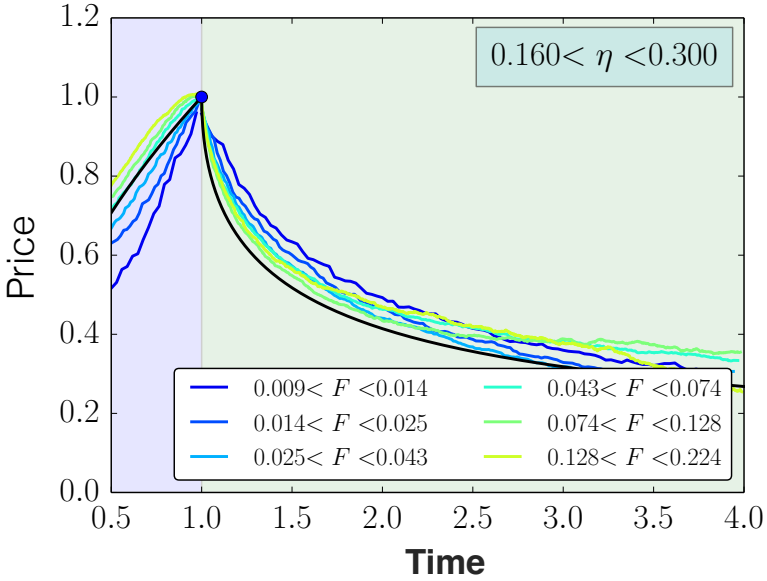
Metaorder execution

A metaorder is a sequence of trades (**child orders**) following a single investment decision.

It is fundamental to evaluate the impact of the orders and, consequently, transaction costs.

Empirically, on proprietary data, price dynamics is concave during the execution and convex after [4].





Can we estimate metaorder impact also with public data?

[4] E. Zarinelli, M. Treccani, J D. Farmer, and F. Lillo, Market Microstructure and Liquidity, 1(02):1550004, 2015.

Data

We used data obtained by LOBSTER [3]

It contains NASDAQ-listed stocks with a precision at least milliseconds up to nanoseconds.

We consider MSFT and AMZN during the 22 training days of June 2021.

We only considered market orders.

Training on 80% of data, Test on the remaining 20%.

We removed the first and last 30 minutes (to eliminate auction effect) and merged orders at the same timestamp.

We want to calibrate the model on these data and then simulate the metaorder dynamics.

[3] lobsterdata.com

Closeness to instability

Fit Hasbrouck model on MSFT and AMZN June 2021.

Cumulative effects of coefficients a_i, c_i negligible with respect the ones of b_i, d_i.

H

$$\begin{cases} r_t = \sum_{i=1}^{p} a_i r_{t-i} + \sum_{i=0}^{p} b_i v_{t-i} + u_{1,t}, \\ v_t = \sum_{i=1}^{p} c_i r_{t-i} + \sum_{i=1}^{p} d_i v_{t-i} + u_{2,t} \end{cases}$$

Good approximation →

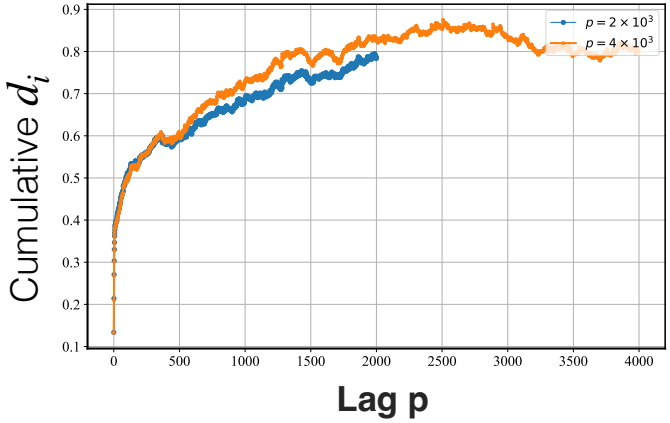
$$\begin{cases} r_t = \sum_{i=0}^{p} b_i v_{t-i} + u_{1,t}, \\ v_t = \sum_{i=1}^{p} d_i v_{t-i} + u_{2,t} \end{cases}$$

TIM

We consider large lags, $p = \{2 \times 10^3, 4 \times 10^3\}$.

The system appears to be **close to non-stationarity** (critical condition)

$$\sum_{i=1}^{p} d_i \to 1, \text{ for } p \to \infty$$



Metaorder simulations

Once the model is calibrated, we **simulate metaorder dynamics inserting child orders δ_v** as you would do in a LOB simulator.

The presence of the autoregressive term seems to be relevant in reproducing stylized facts.

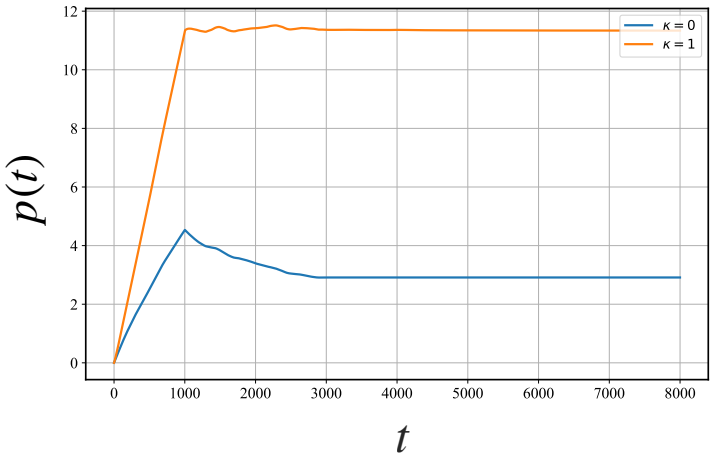
$$\begin{cases} r_t = \sum_{i=0}^{p} b_i v_{t-i} \\ v_t = \kappa\left(\sum_{i=1}^{p} d_i v_{t-i}\right) + \delta_v \end{cases}$$

Child Metaorder volume

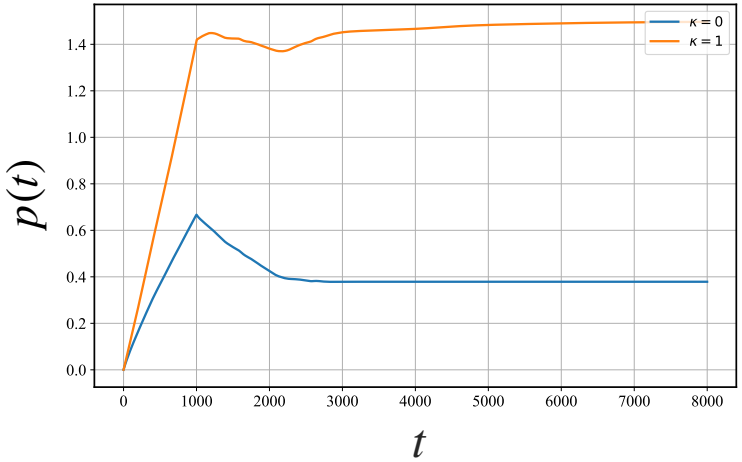
$$\kappa = \{0,1\}$$

For $\kappa = 0$ the child orders only affects prices.

Amazon



Microsoft



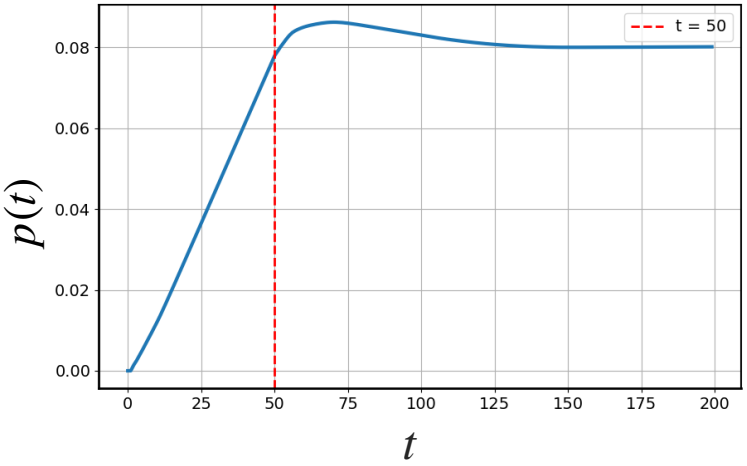
Metaorder simulations

May the problem be linearity?

Also highly non linear models fail to reproduce this stylized facts.

The growth is linear and we do not observe the price inversion.

LSTM model on MSFT



More results in **M.Naviglio, F.Lillo, in preparation.**

Similar results observed with linear[5] models and GANs[6].

[5] Salma Elomari-Kessab, Guillaume Maitrier, Julius Bonart, and Jean-Philippe Bouchaud, arXiv:2405.10654, 2024.
[6] Rama Cont, Mihai Cucuringu, Jonathan Kochems, and Felix Prenzel. SSRN 4512356, 2023.

What we learn from simulations

- The coefficients d_i have large cumulative contribution and they are close to non stationarity as the number of lags increase;
- This means that volumes dynamics tend to have long memory. Thus, the effect of a child order could be very persistent;

What is happening with $\kappa = 1$?

- Cumulative effects on volumes also influence price dynamics. Turning off the autoregressive term in volumes we find back the expected concave and convex pattern.
- The volume dynamics triggered by the shock δ_v generates large effects in prices.

This suggests that child orders should either not be included in volume dynamics at all, or only partially included.

A modified TIM model

We propose a modified TIM model to describe metaorder dynamics

Metaorder V executed in $[0,T]$ (θ is the Heaviside function)

$V = \delta_v$ for $\alpha = 1$.

$$p(t) = \int_0^t G(t-\tau)\left[v(\tau) + (1-\alpha)V\theta(T-\tau)\right]d\tau$$

$$v(t) = \alpha V\theta(T-t) + \lambda \int_0^t \mathscr{D}(t-\tau)v(\tau)d\tau$$

$\mathscr{D}$, G are given kernel functions.

$\alpha \in [0,1]$ weights child orders contribution in volumes and prices.
Small α suppresses in prices the part coming from volumes dynamics.

Volume's equation is a Volterra Equation of the first type solvable as

$$v(t) = \mathscr{L}^{-1}\left[\frac{\mathscr{L}[f]}{1-\lambda\mathscr{L}[\mathscr{D}]}\right](t)$$

$\mathscr{L}$ is the Laplace transform and $f = \alpha V\theta(T-t)$.

Analytical for $\mathscr{D} = e^{-\beta t}$, $G = e^{-\rho t}$. The stationary condition is;

$$\lambda \int_0^\infty e^{-\beta t}\, dt < 1 \;\rightarrow\; \beta > \lambda$$

Price solution during the metaorder

During the metaorder the price solution reads

$$p(t<T)=\begin{cases}\alpha V\left[\frac{\beta}{\beta-\lambda}\frac{1-e^{-\rho t}}{\rho}+\left(1-\frac{\beta}{\beta-\lambda}\right)\frac{e^{(-\beta+\lambda)t}-e^{-\rho t}}{\rho-\beta+\lambda}\right]+(1-\alpha)V\frac{1-e^{-\rho t}}{\rho}, & \beta>\lambda\\ \alpha V\left[\frac{1-e^{-\rho t}}{\rho}+\beta\frac{\rho t+e^{-\rho t}-1}{\rho^2}\right]+(1-\alpha)V\frac{1-e^{-\rho t}}{\rho}, & \beta=\lambda\end{cases}$$

while after

$$P(t\geq T)=\alpha V\left\{\frac{\beta}{\beta-\lambda}\frac{e^{-\rho(t-T)}-e^{-\rho t}}{\rho}+\left(1-\frac{\beta}{\beta-\lambda}\right)\left[\frac{e^{-\rho(t-T)}-e^{-\rho t}}{\rho-\beta+\lambda}+\frac{e^{(-\beta+\lambda)t}-e^{-(\beta-\lambda)(t-T)}}{\rho-\beta+\lambda}\right]\right\}+(1-\alpha)V\frac{e^{-\rho(t-T)}-e^{-\rho t}}{\rho} \quad \beta>\lambda$$

$$P(t\geq T)=\alpha V\left\{\frac{e^{-\rho(t-T)}-e^{-\rho t}}{\rho}+\beta\frac{\rho T+e^{-\rho t}-e^{-\rho(t-T)}}{\rho^2}\right\}+(1-\alpha)V\frac{e^{-\rho(t-T)}-e^{-\rho t}}{\rho} \quad \beta=\lambda$$

The part proportional to α is inherited from volume dynamics.

For $\alpha=0$ the growth is concave. For $\alpha=1$, a linear term survives during metaorder.

After metaorder execution price tends to a nonzero value.

If $\alpha>0$ and $\beta\simeq\lambda$ prices have permanent impact even with exponential kernels.

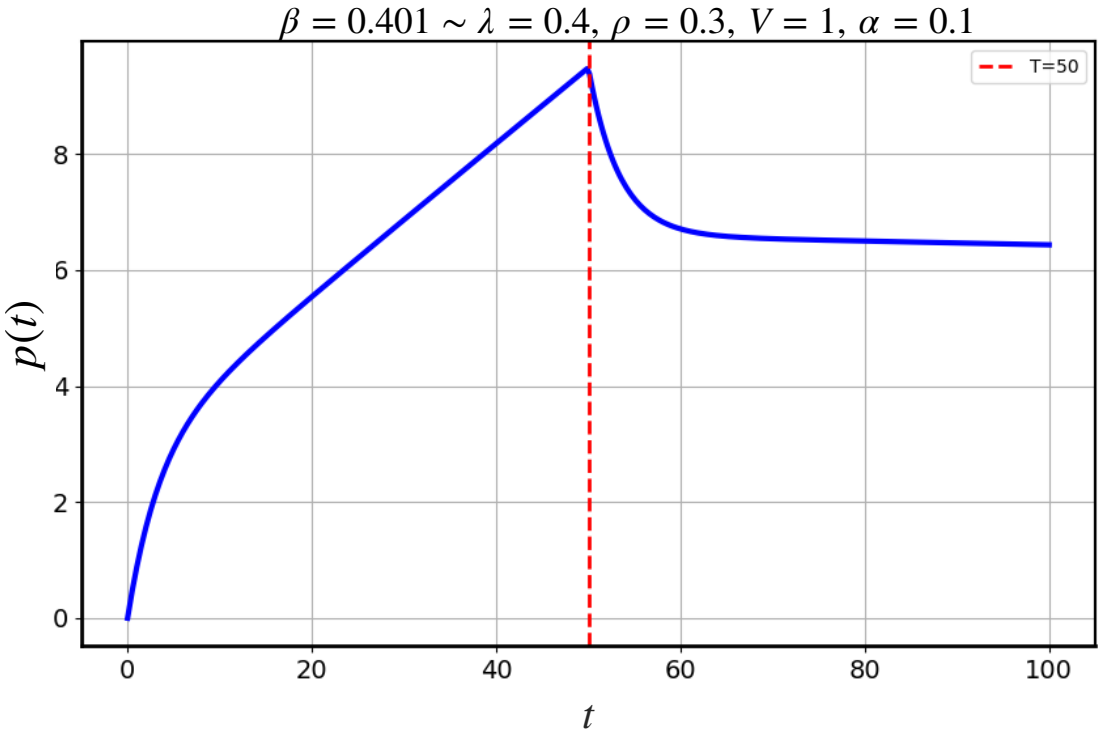
Tuning α

Even with short range kernels it is possibile to observe <u>permanent impact</u> close to criticality. But, volumes contribution in prices must be properly rescaled with α.

This is true even with very small values of α.

Prices are concave during the execution and convex after.

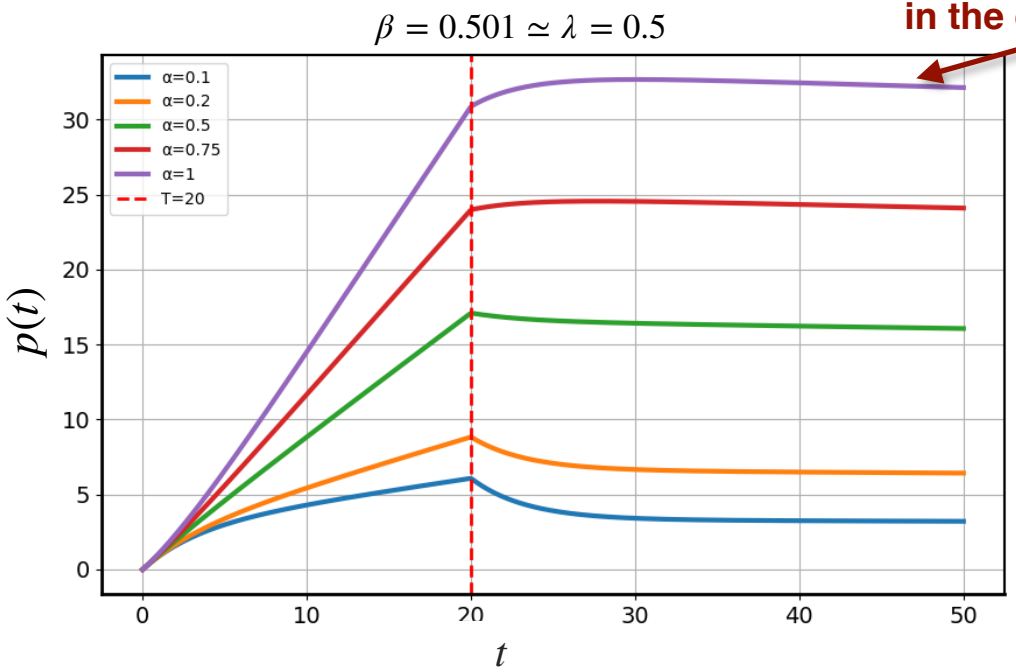
At the end of the metaorder, when exponential kernels go to zero, prices increase linearly.



The role of α

Prices inherit a linear term during the execution and a constant one after.
Both proportional to α.

Childs order only in the order flow



The role of α is to suppress non-concave contributions and long impact effects coming from volumes.

This allows to reproduce stylized facts on metaorders, even close to criticality.

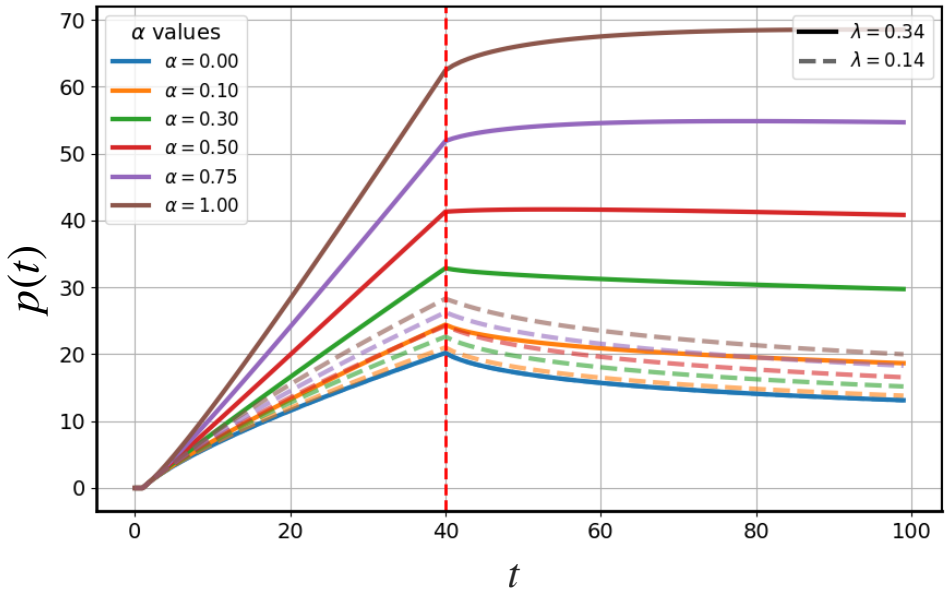
To simulate metaorder dynamics child orders contribution in volumes must be suppressed.
Particularly relevant in LOB simulators.

Discrete time simulations

Real data are more consistent with power law kernels

In this case the model is not analytically solvable.

We simulate the correspondent discrete model.



$$p_t = \sum_{i=1}^{t} g_i[v_{t-i} + (1-\alpha)V\theta(T-(t-i))]$$

$$v_t = \alpha V\theta(T-t) + \lambda \sum_{i=1}^{t} d_i v_{t-i}$$

$$g_i = \frac{1}{i^\eta},\ d_i = \frac{1}{i^\delta} \text{ with } \eta = 0.25,\ \delta = 1.5.$$

Going farther from criticality the solutions (dashed) become similar.

Conclusions

- Fitting the model on data, makes volume dynamics very close to non stationarity. Simulating child orders on volumes generates large impact effects even on prices. This is relevant especially in LOB simulators.

- We proposed a modified TIM model incorporating metaorder dynamics. We show that only a small portion of child orders should affect order flow.
  Furthermore, **even if a small portion of child orders** affects order flow under critical conditions **permanent impact on prices is generated**.
  In LOB simulators, when metaorder are simulated the effects on order flow should be suppressed.

- The **LMF model**[7] suggests that volume autocorrelations arise from independent simultaneous metaorders. The addition of a further metaorder does not modify order flow. Models correctly capture the correlations, they miss the causality.

- In the future: it would be interesting to estimate α on real data.

[7] Fabrizio Lillo, Szabolcs Mike, and J Doyne Farmer. Physical Review E—Statistical, Nonlinear, and Soft Matter Physics, 71(6):066122, 2005.

Backup slides

Price and Trade Models

A general price and trade model can be written as

$$r_t = \mathcal{G}_1(r_{t-1}, r_{t-2}, \ldots, v_t, v_{t-1}, v_{t-2}, \ldots)$$

$$v_t = \mathcal{G}_2(r_{t-1}, r_{t-2}, \ldots, v_{t-1}, v_{t-2}, \ldots)$$

where $r = \log\left(p_t / p_{t-1}\right)$ are the log returns.

Time advances by one at each transaction.

The model is described by the functions $\mathcal{G}_1$ and $\mathcal{G}_2$.

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Price and Trade Models

A general price and trade model describes their joint dynamics, possibly by taking into account other covariates

$$p_t = \mathscr{F}_1(p_{t-1}, p_{t-2}, \ldots, v_t, v_{t-1}, v_{t-2}, \ldots)$$

$$v_t = \mathscr{F}_2(p_{t-1}, p_{t-2}, \ldots, v_{t-1}, v_{t-2}, \ldots)$$

where p is the **price** just the trade and **v** is the **signed volume**. (transaction time)

$\mathscr{F}_1$ and $\mathscr{F}_2$ define the **model** which describes the relationship between the present quantities with the lagged ones.

Non Parametric and Non Linear Models: Neural Networks

A NN is a highly nonlinear function f_θ mapping input $\mathcal{X}$ to output x_t, with parameters θ learned from data trough the optimization of a loss function (e.g. MSE).

We refer to p as **lag order** in analogy with the VAR case. It is the length of the input time series.

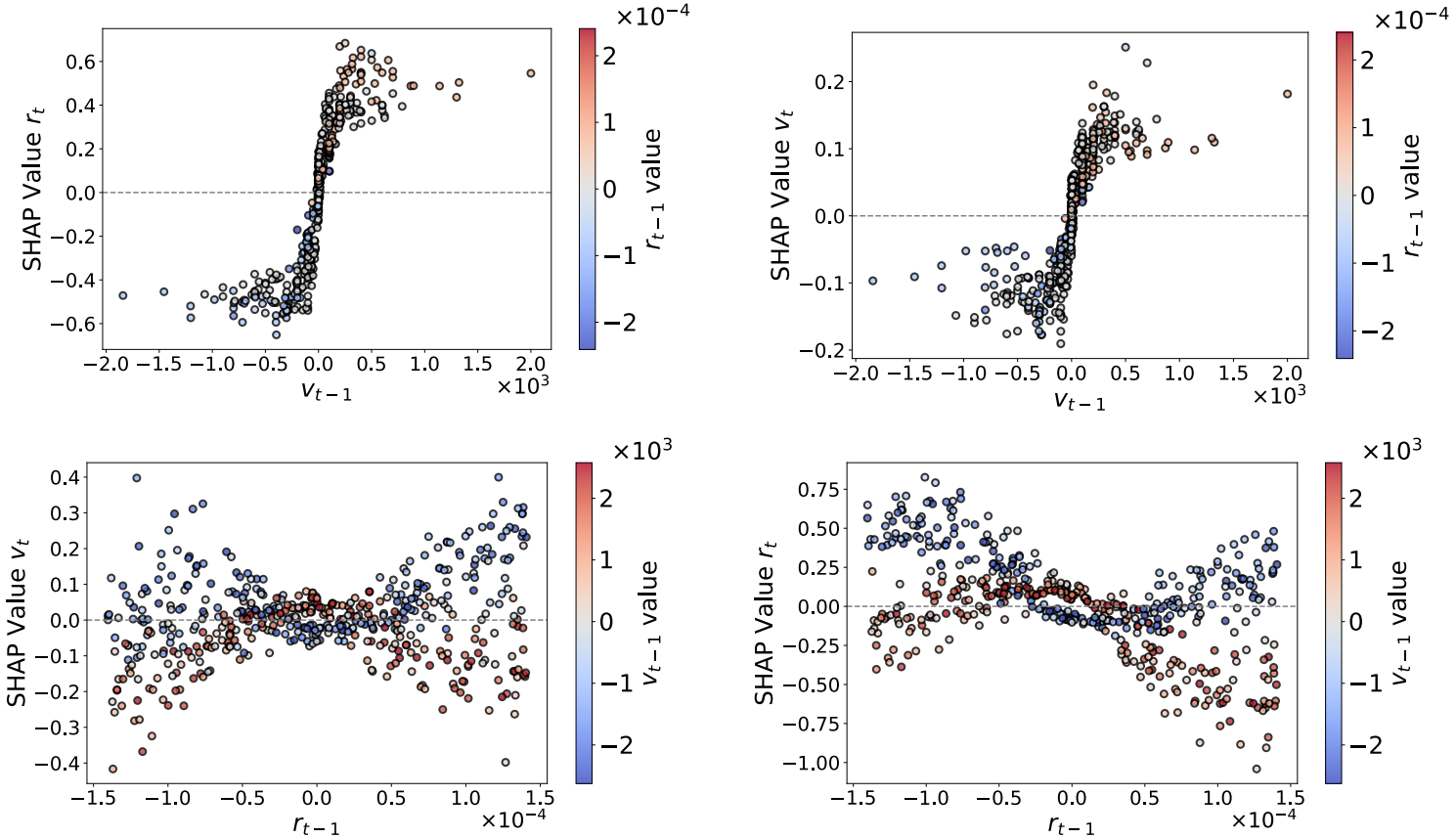
In our work f is built as

- Long Short Time Memory (LSTM) layers;
- Convolutional (CNN) layers;
- Dense layers.

$+$ $\tanh$/**sigmoid** activation functions.

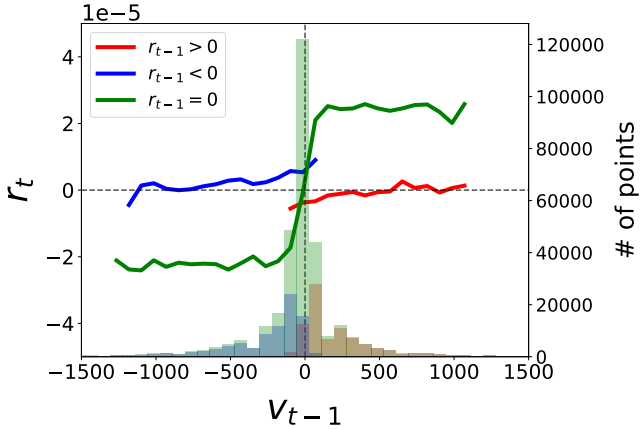
SCUOLA NORMALE SUPERIORE

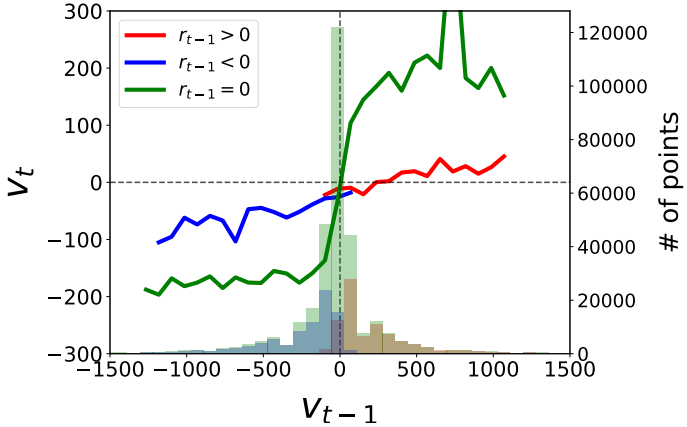
Non linear relations between variables

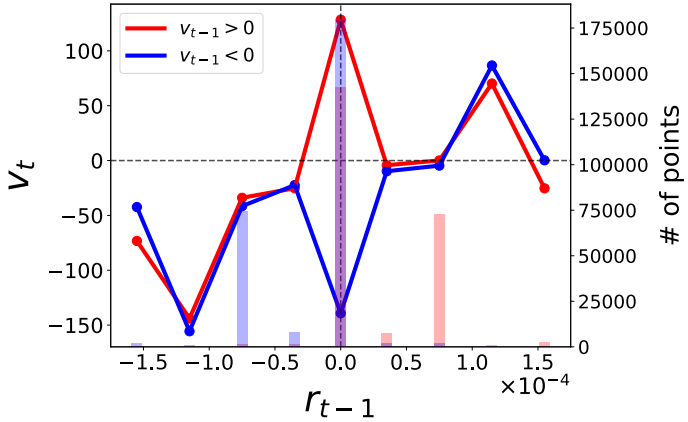


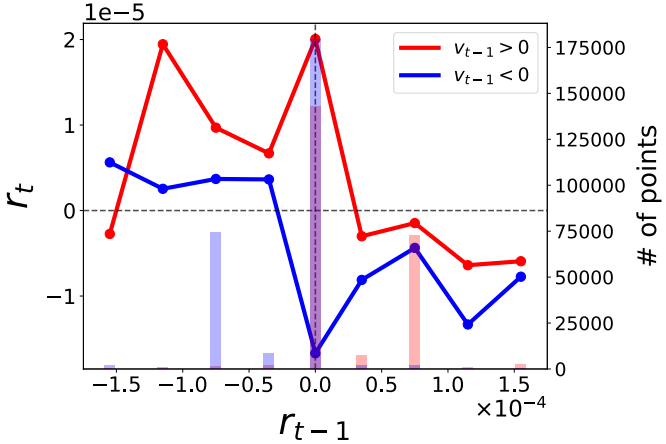
SCUOLA NORMALE SUPERIORE

Non linear relations between variables in data



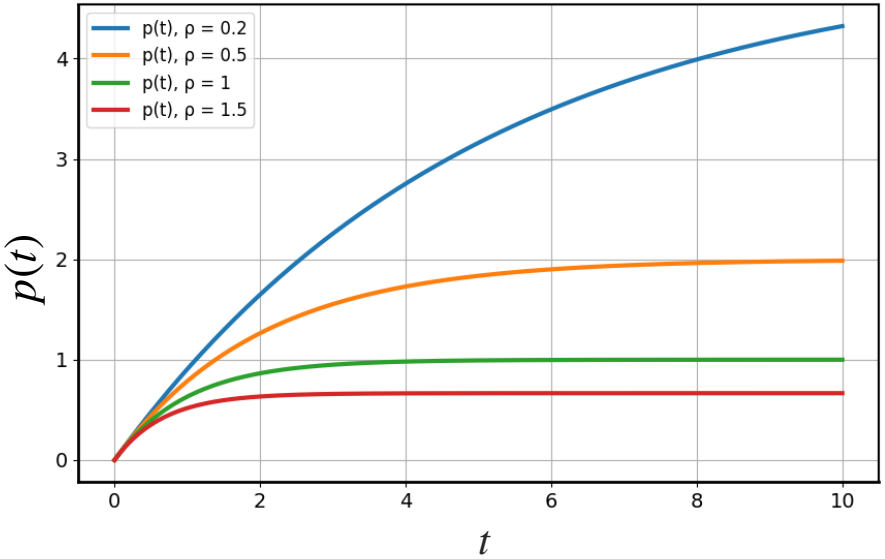




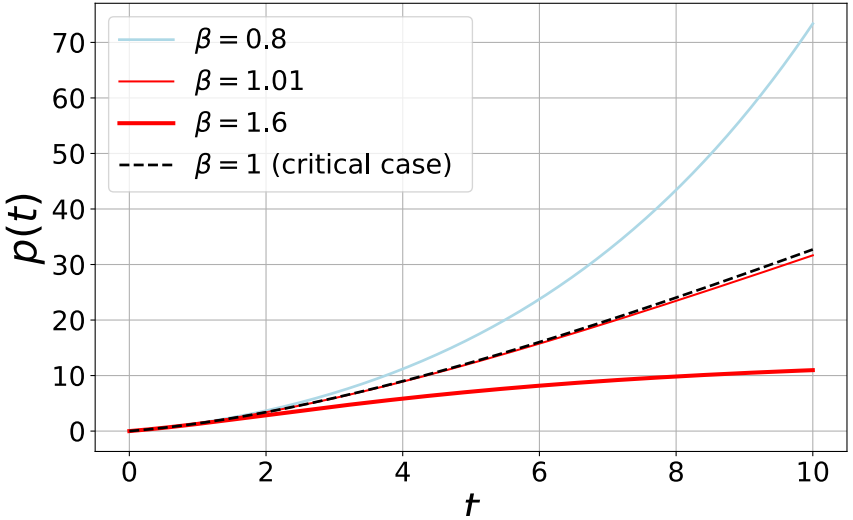


Price solution during the metaorder

$\alpha = 0$



$\alpha = 1,\ \lambda = 1$



In the critical case, a linear term survives. Decreasing α we suppress this contribution.

Volume solution

Analytically solvable for $\mathscr{D} = e^{-\beta t}$, $G = e^{-\rho t}$.

Stationary condition for volumes in the continuum reads

Equivalent to $\sum_{i=1}^{\infty} d_i < 1$, in discrete time

$$\lambda \int_0^{\infty} e^{-\beta t}\, dt < 1 \;\rightarrow\; \beta > \lambda$$

The volume solution reads

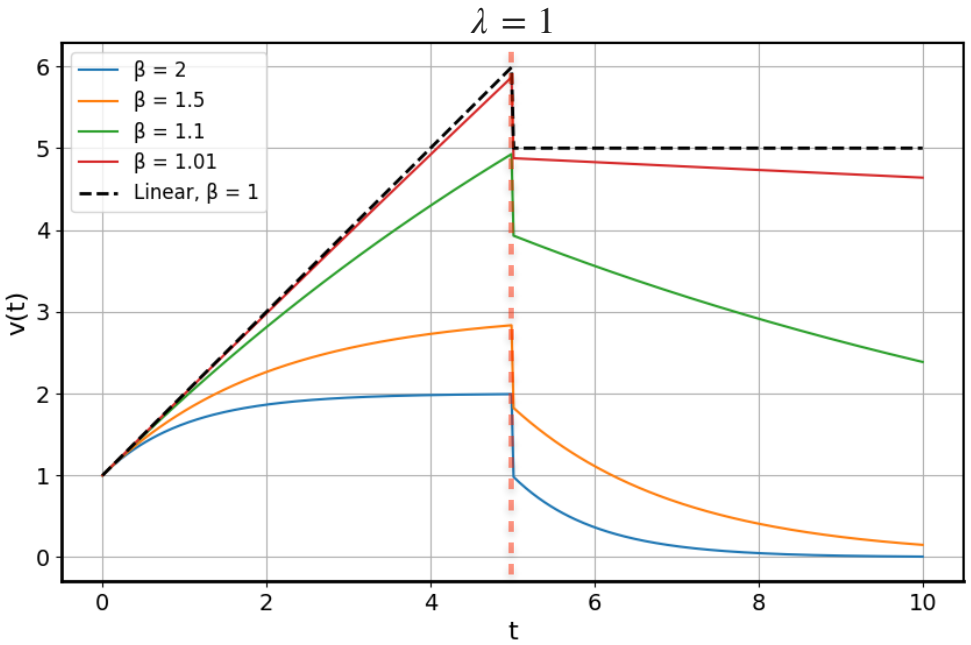
$$v(t) = \begin{cases} \alpha V \left\{ \left(\frac{\beta}{\beta-\lambda} + \left(1 - \frac{\beta}{\beta-\lambda}\right) e^{-(\beta-\lambda)t} \right) - \theta(t-T) \left(\frac{\beta}{\beta-\lambda} + \left(1 - \frac{\beta}{\beta-\lambda}\right) e^{-(\beta-\lambda)(t-T)} \right) \right\}, & \text{if } \beta \neq \lambda, \\ \alpha V \left\{ \underline{1 + \beta t} - \theta(t-T)[1 + \beta(t-T)] \right\}, & \text{if } \beta = \lambda. \end{cases}$$

Volume solution is proportional to α.

As we get closer to the critical case, $\beta = \lambda$, the volume tends to be linear during the metaorder and constant after.

Volume solution

Volume contribution is proportional to α.



- $\beta > \lambda$: concave growth during the metaorder and convex relaxation to zero.
- $\beta = \lambda$: linear growth during the metaorder and instantaneous stabilization at $t = T$:

$$v(t > T) = \alpha V \beta T$$

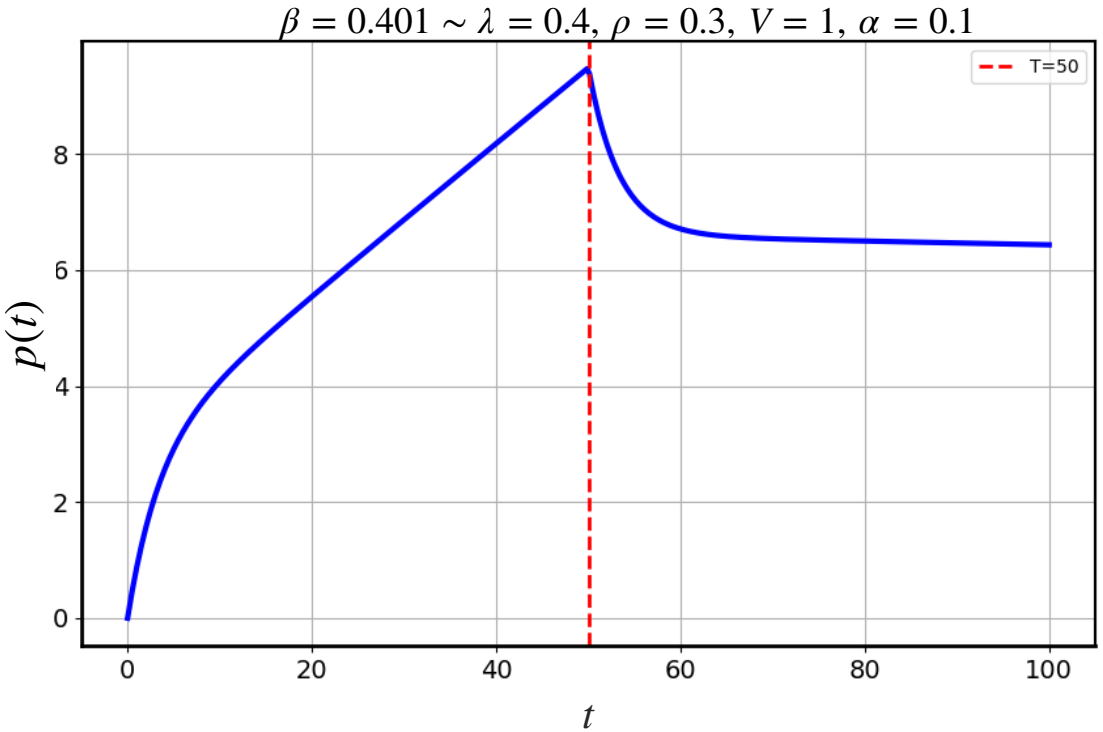
Permanent impact on volumes is possible even with exponential kernels.

If the system is close to criticality ($\beta \gtrsim \lambda$) volumes relax on very large timescales.

Tuning α

Even with short range kernels it is possibile to observe <u>permanent impact</u> close to criticality. But, volumes contribution in prices must be properly rescaled with α.

This is true even with very small values of α.

Prices are concave during the execution and convex after.

At the end of the metaorder, when exponential kernels go to zero, prices increase linearly.